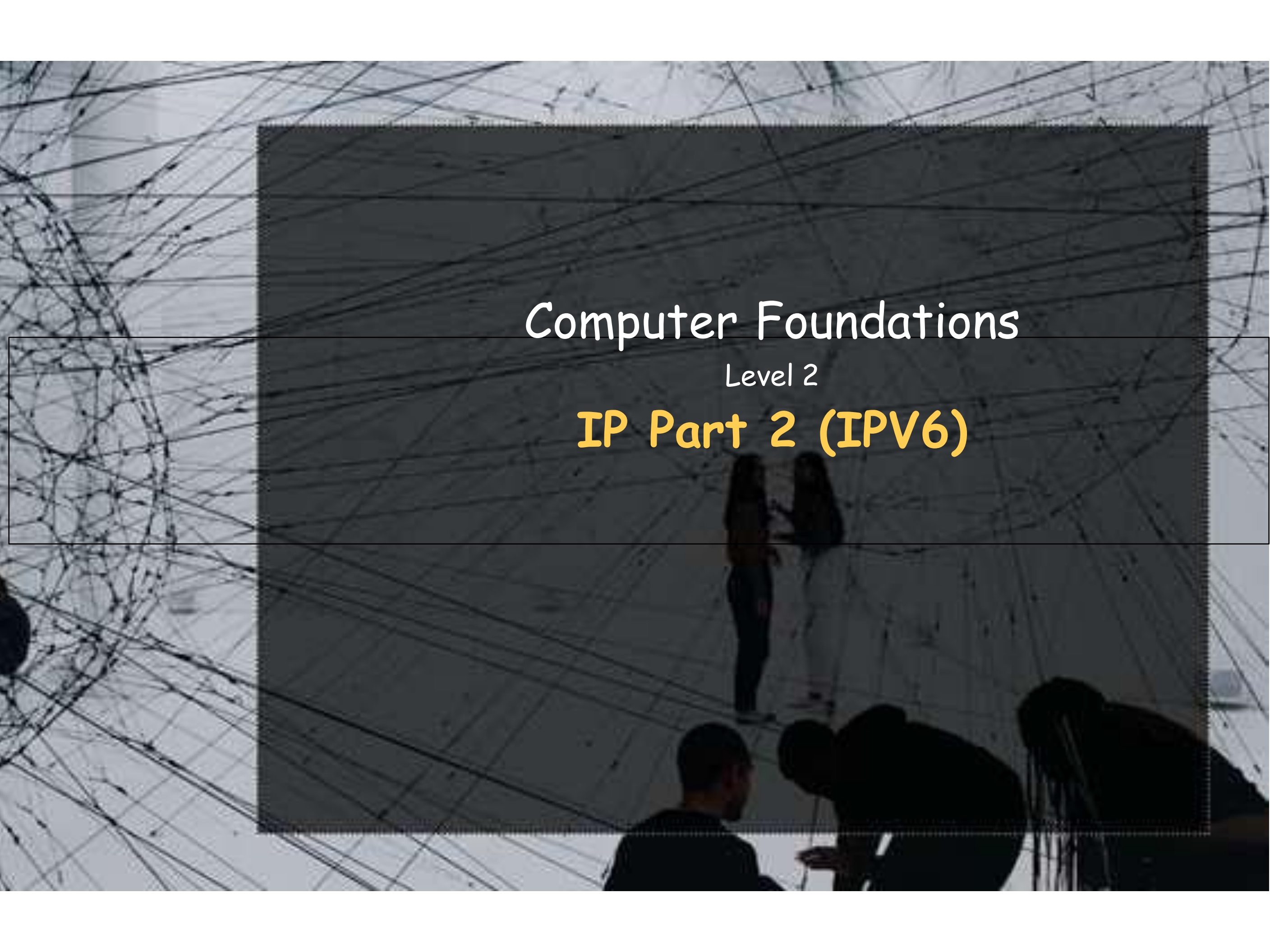




Computer Foundations Level 2



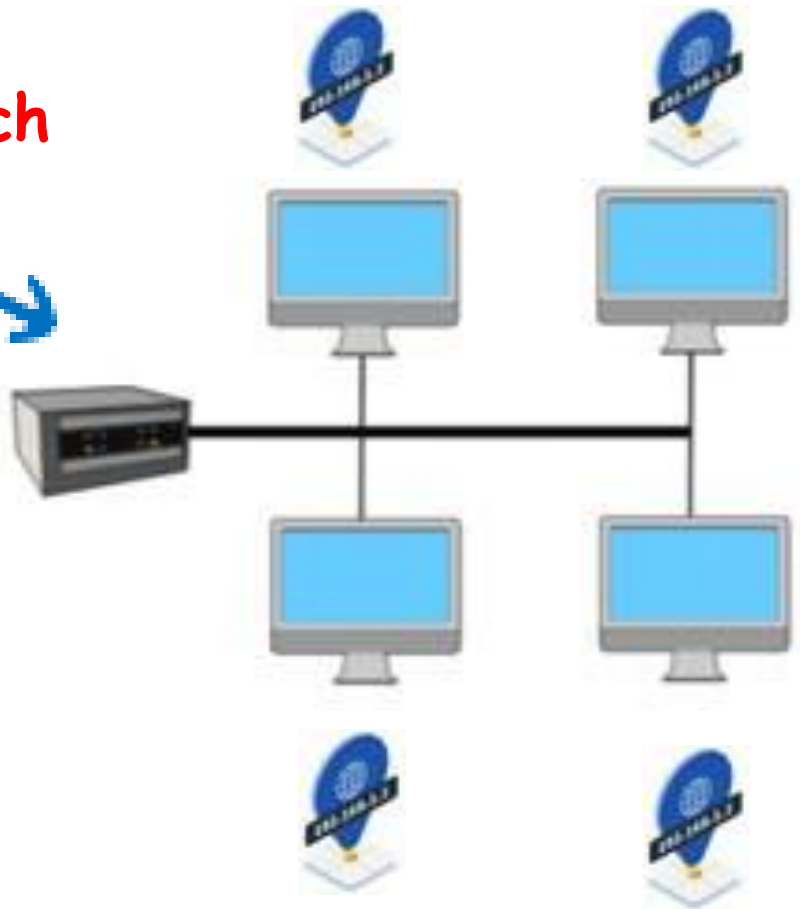
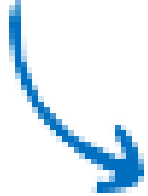
Computer Foundations

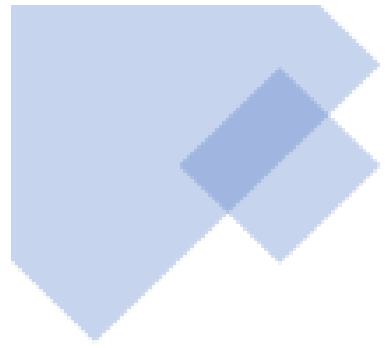
Level 2

IP Part 2 (IPV6)

How to locate certain device?

Switch



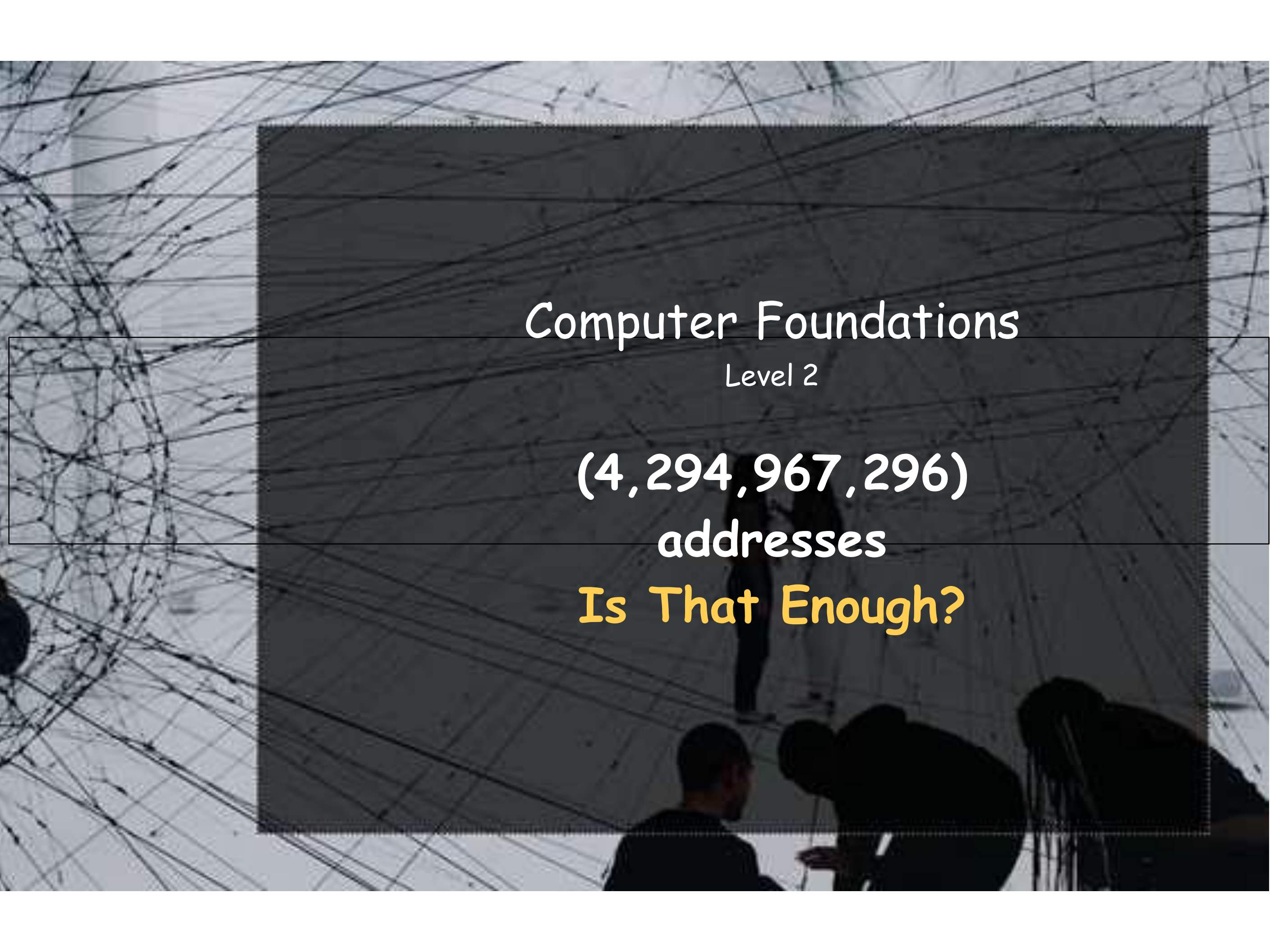


IPv4 Address Format (Dotted Decimal Notation)



provides 2³² (4,294,967,296) addresses





Computer Foundations

Level 2

(4,294,967,296)

addresses

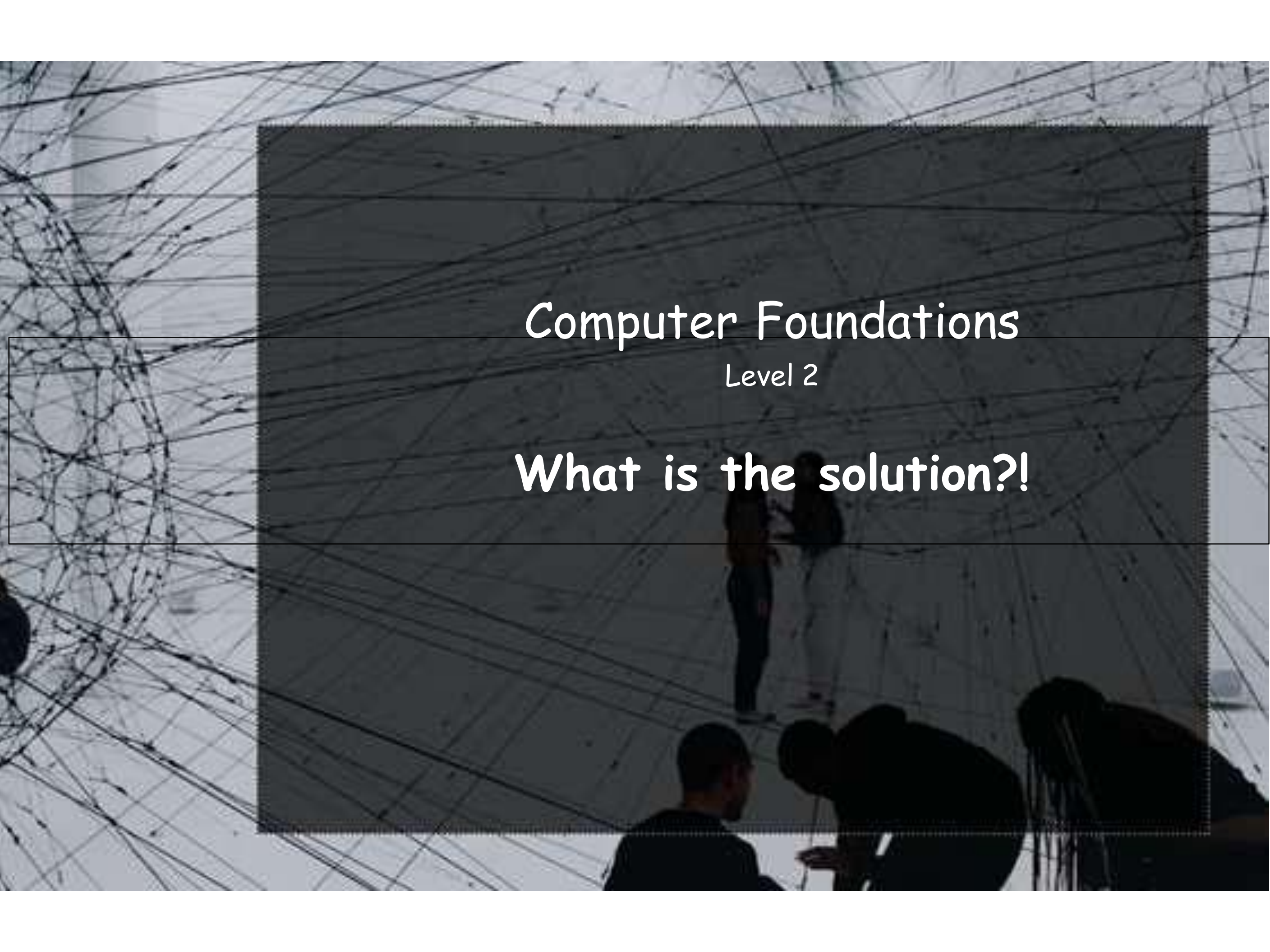
Is That Enough?

Computer Foundations

Level 2

(4,294,967,296)
addresses

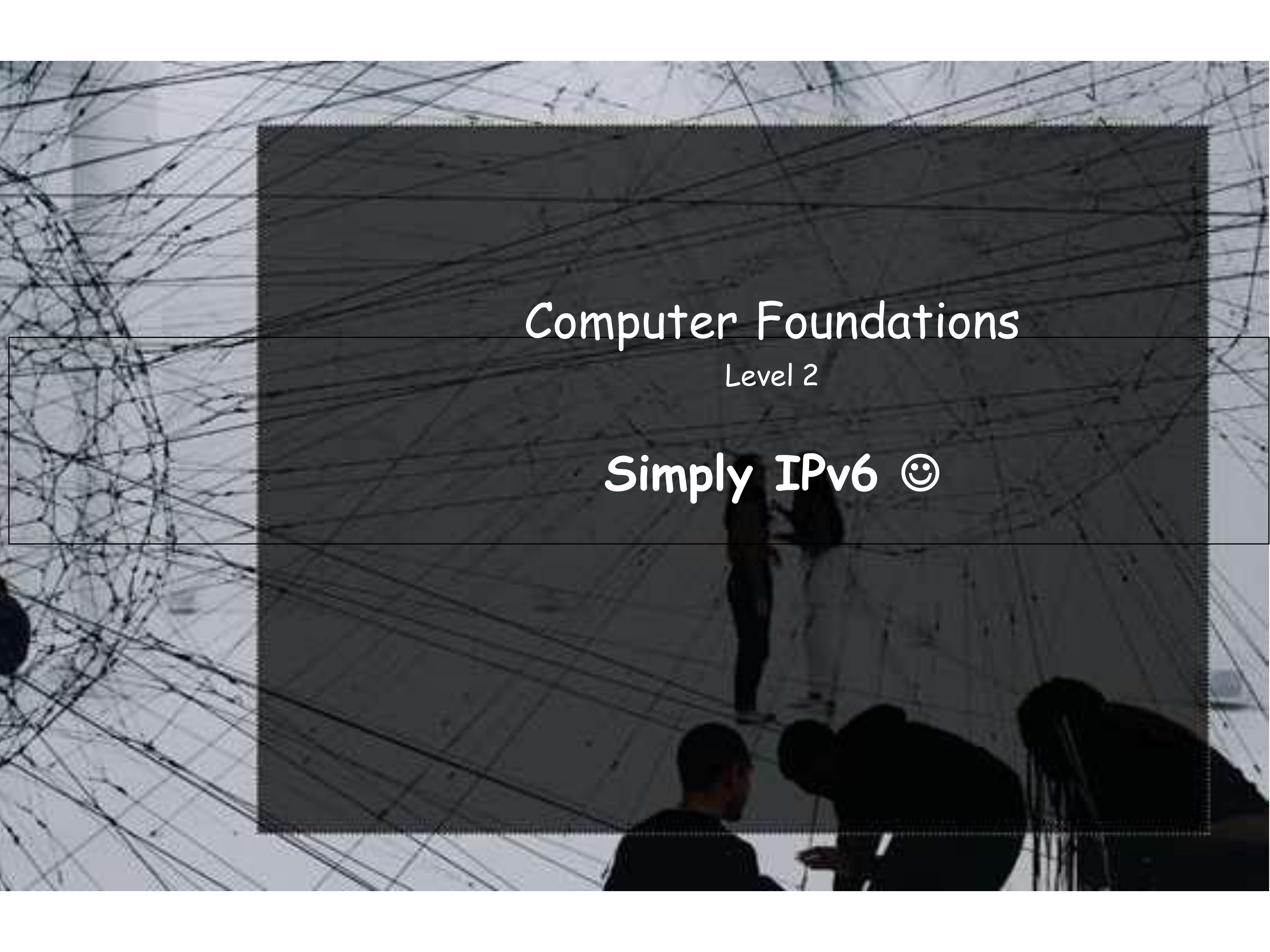
It Was Enough, now
NO NO NO

The background of the slide features a complex, abstract network of thin, dark lines crisscrossing a light gray background. In the lower portion of the image, there are dark silhouettes of several people, some standing and some leaning over, suggesting a collaborative work environment or a meeting. A large, dark gray rectangular area is overlaid on the right side of the image, serving as a backdrop for the text.

Computer Foundations

Level 2

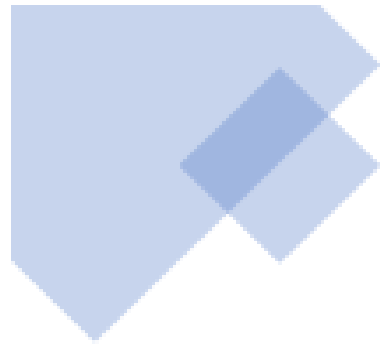
What is the solution?!



Computer Foundations

Level 2

Simply IPv6 😊

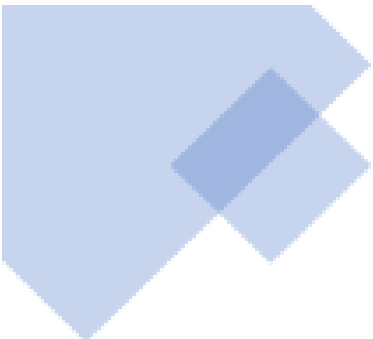


IPv4 Address Format (Dotted Decimal Notation)



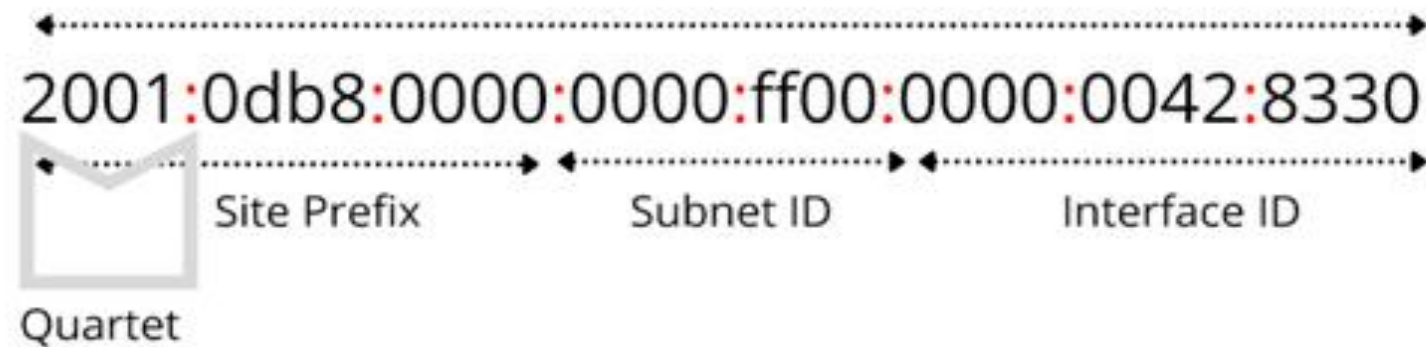
provides 2³² (4,294,967,296) addresses





IPv6 Address Format (Hexadecimal Notation)

16 Bytes = 128 Bits

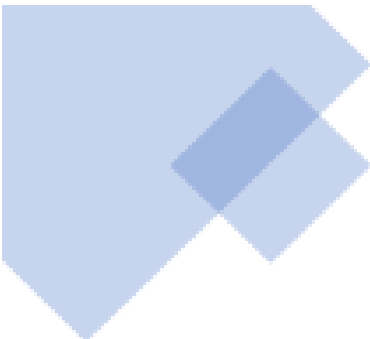


Omit any leading zeros within a quartet. `2001:db8::ff00:0:42:8330`

Short version of ipv6

provides 232 (4,294,967,296) addresses





IP settings

IP assignment:

Automatic (DHCP)

Edit

Properties

SSID: Mohammed's Galaxy S22 Ultra

Protocol: Wi-Fi 4 (802.11n)

Security type: WPA2-Personal

Network band: 2.4 GHz

Network channel: 6

Link speed (Receive/Transmit): 1300/144 (Mbps)

Link-local IPv6 address: fe80::708e:8e15:8a48:57f4%15

IPv4 address: 192.168.151.153

IPv4 DNS servers: 192.168.151.168

Manufacturer: Broadcom

Description: Broadcom 802.11ac Network Adapter

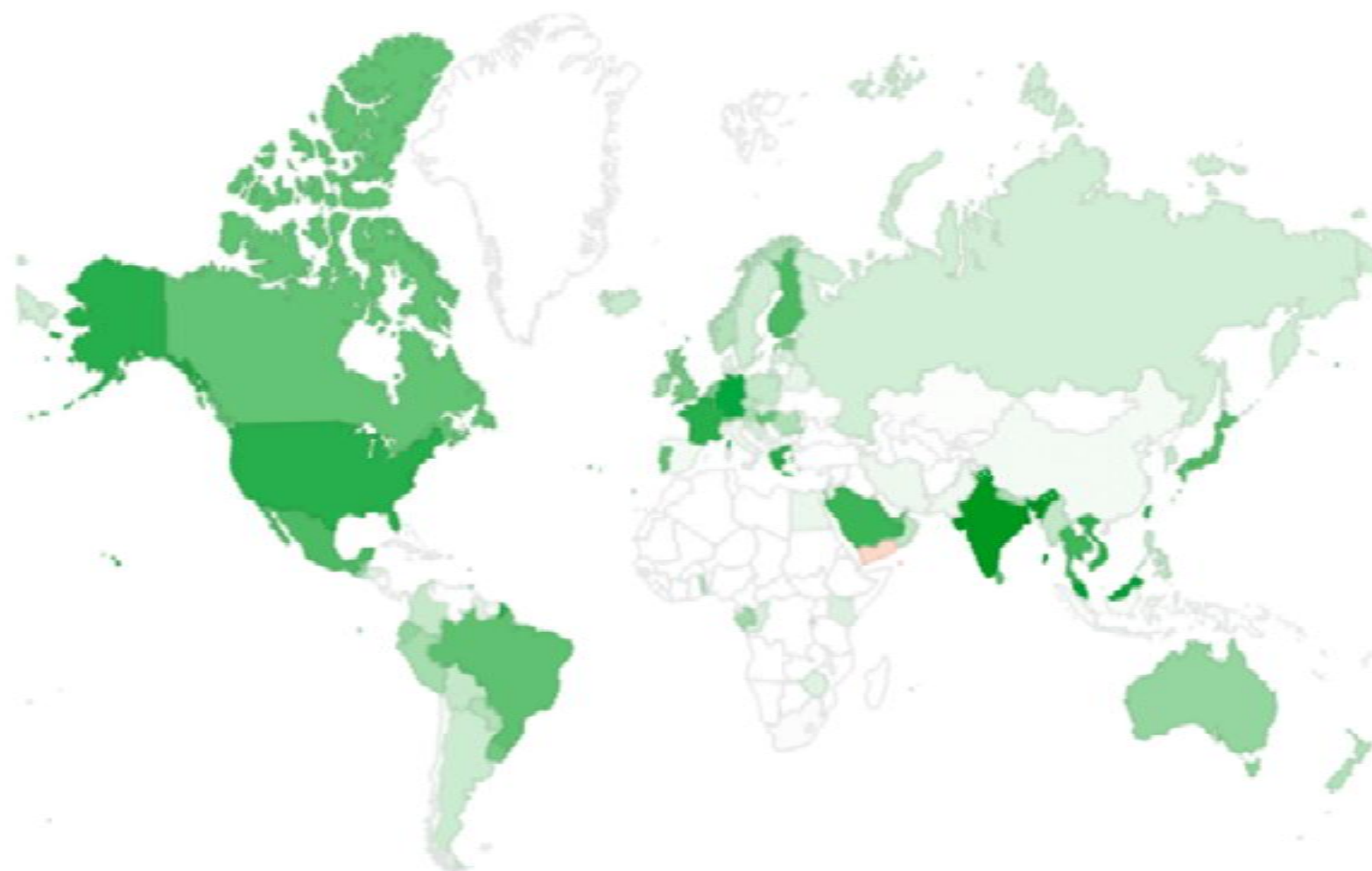
Driver version: 101.10.603.0

Physical address (MAC): 88-66-5A-03-A3-AB

Copy



Per-Country IPv6 adoption



Thank you 😊